Juan M. Haut

Associate Professor (Profesor Titular de Universidad)

Department of Computer and Communication Technology
University of Extremadura | Hyperspectral Computing Laboratory (HyperComp)

juanmariohaut@unex.es | Website | Google Scholar | ORCID | GitHub

Cáceres, Spain | ORCID: 0000-0001-6701-961X

Associate Professor (Profesor Titular de Universidad) at the Department of Computer and Communication Technology, University of Extremadura, and member of the Hyperspectral Computing Laboratory (HyperComp). Research focused on deep learning for hyperspectral remote sensing, GPU acceleration and distributed computing, with an emphasis on computational efficiency. IEEE Senior Member, Chair of the Spanish Chapter of IEEE GRSS, Associate Editor of Remote Sensing, and Editorial Board Member of Scientific Reports.

ACADEMIC APPOINTMENTS

2026 - present **Profesor Titular de Universidad**

University of Extremadura | Cáceres, Spain

Architecture and Computer Technology area. Research in deep learning, hyperspectral remote sensing and high-performance computing. Teaching Arquitecturas Especializadas, Sistemas Multimedia and Computación Paralela in 2026-2027.

Appointment published in the BOE on 23 May 2026

2023 - 2026 **Profesor Contratado Doctor**

University of Extremadura | Cáceres, Spain

Architecture and Computer Technology area. Teaching Arquitecturas Especializadas, Diseño de Sistemas Operativos, Tecnología de Computadores.

2021 - 2023 **Profesor Ayudante Doctor**

University of Extremadura | Cáceres, Spain

2020 - 2021 **Profesor Ayudante Doctor**

Universidad Nacional de Educación a Distancia (UNED) | Madrid, Spain

2019 **Profesor Sustituto**

University of Extremadura | Cáceres, Spain

2015 - 2019 **Predoctoral Researcher (FPU Grant)**

University of Extremadura | Cáceres, Spain

EDUCATION

2015 - 2019 **PhD in Information Technology** | University of Extremadura

International Mention; Sobresaliente Cum Laude; Outstanding Doctoral Award.
Supervisors: Antonio Plaza and Javier Plaza.

2014 - 2016 **MSc in Computer Engineering** | University of Extremadura

2013 - 2014 **MSc in Secondary Education Teaching** | University of Extremadura

2010 - 2014 **BSc in Computer Engineering** | University of Extremadura

2007 - 2011 **BSc in Computer Management Engineering** | University of Extremadura

RESEARCH LEADERSHIP

OASIS - Optimización de la IA para Iniciativas Sostenibles de Teledetección Hiperespectral

National project funded by Agencia Estatal de Investigación. PI: J.M. Haut. 163,750 €.

Deep Learning optimization for sustainable hyperspectral remote sensing; GPU acceleration and energy efficiency

2024-09-01 to 2027-12-31

RAT_EOS_PC - Early Warning Networks for Climate Risk Remote Sensing

European Commission project. PI: J.M. Haut. 548,502 €.

Earth observation satellites for civil protection; Climate change risk detection

2023-09-01 to 2026-12-31

PROFESSIONAL SERVICE

Chair of the Spanish Chapter of IEEE GRSS.

Associate Editor of Remote Sensing since 2026.

Editorial Board Member of Scientific Reports.

Guest Editor of multiple special issues, including Cutting-Edge Algorithms and Characteristic Geo-Applications with Remote Sensing.

Editorial service has included Topical Editor for IEEE TGRS and Associate Editor for IEEE GRSL and IEEE J-MASS, as documented in the University of Extremadura's 2022 research profile.

AWARDS AND DISTINCTIONS

2022 **Research Excellence Award (Young Researchers, technical field)** | UEx

2021 **Outstanding Paper Award - IEEE HyperMLPA Workshop** | IEEE

2021 **Best Reviewer - IEEE Trans. Geoscience and Remote Sensing** | IEEE GRSS

2020 **Best PhD Thesis in Remote Sensing** | IEEE Spain GRSS Chapter

2020 **Outstanding Doctoral Award (Premio Extraordinario de Doctorado)** | UEx

2019 **Outstanding Paper Award - IEEE WHISPERS** | IEEE

2018 **Best Reviewer - IEEE Geoscience and Remote Sensing Letters** | IEEE GRSS

TECHNICAL EXPERTISE

Deep Learning: CNNs, RNNs, GANs, Transformers, Capsule Networks, Neural ODEs, Foundation Models

High-Performance Computing: GPU Programming, CUDA, Triton, Distributed Computing, Cloud Computing, Apache Spark

Remote Sensing: Hyperspectral Imaging, Image Classification, Anomaly Detection, Unmixing, Super-Resolution

Programming: Python, PyTorch, TensorFlow, C/C++, CUDA, Docker, Linux

ACCREDITATIONS AND LANGUAGES

2023-04-20 | **Profesor Titular de Universidad** | ANECA

2019-11-25 | **Profesor Contratado Doctor** | ANECA

2019-11-25 | **Profesor Ayudante Doctor** | ANECA

2024 | **Sexenio CNEAI (2016-2021)** | ANECA / CNEAI

Spanish: Native speaker | **English**: B1 | **Portuguese**: A2

SELECTED JOURNAL ARTICLES

A selection of 8 journal articles illustrating recent methods, applications and foundational work. Complete publications and current citation metrics: publication list and Google Scholar.

01. 3D-SCube: Information-preserving spatial-spectral rearrangement for hyperspectral semantic segmentation
M. B. García-Flores, C. Cañada-Rostro, M. E. Paoletti, **J. M. Haut**
Expert Systems with Applications, 333, 133926 (2027). Journal volume dated January 2027.
doi:10.1016/j.eswa.2026.133926

02. DRAGON-BS: A Unified Gradient-Boosted Reinforcement Learning Framework for Interpretable Band Selection
M. B. García-Flores, C. Cañada-Rostro, M. E. Paoletti, **J. M. Haut**
IEEE Transactions on Geoscience and Remote Sensing (2026). Early Access.
doi:10.1109/TGRS.2026.3741200

03. Towards knowledge-infused seabed sediment mapping: A semi-supervised framework integrating large language models and knowledge graphs for multibeam data
H. Wang, W. Chen, X. Li, G. Zhou, Q. Liang, J. Li, M. E. Paoletti, **J. M. Haut**
Engineering Applications of Artificial Intelligence, 182, 115757 (2026).
doi:10.1016/j.engappai.2026.115757

04. Scaling and Validating a Distributed Training and Retraining Pipeline for Network Intrusion Detection
J. M. Franco-Valiente, J. Calle-Cancho, A. Serrano-Martín, **J. M. Haut**
Electronics, 15(18), 4074 (2026).
doi:10.3390/electronics15184074

05. Remote Sensing of the Coastal Zone: An ever-growing concern for coastal sustainability
B. Fang, X. Zhang, L. Wang, R. Kou, M. E. Paoletti, **J. M. Haut**, A. Plaza
IEEE Geoscience and Remote Sensing Magazine, 2-40 (2026).
doi:10.1109/MGRS.2026.3716182

06. Condensed and Discriminative Feature Encoding for Remote Sensing Image Retrieval
L. Han, Y. Ge, M. E. Paoletti, S. Moreno-Álvarez, Y. Xu, **J. M. Haut**
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 19, 25537-25550 (2026).
doi:10.1109/JSTARS.2026.3707402

07. Distributed Parallel Hyperspectral Unmixing for Large-Scale Data in Spark Environments via Geometric Distance
C. Cañada, M. E. Paoletti, M. B. García-Flores, X. Tao, R. Pastor-Vargas, **J. M. Haut**
IEEE Transactions on Geoscience and Remote Sensing, 63, 1-19 (2025).
doi:10.1109/TGRS.2025.3624287

08. Deep Learning Classifiers for Hyperspectral Imaging: A Review
M. Paoletti, **J. Haut**, J. Plaza, A. Plaza
ISPRS Journal of Photogrammetry and Remote Sensing, 158, 279-317 (2019).
doi:10.1016/j.isprsjprs.2019.09.006